



BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, Pilani
Pilani Campus
ACADEMIC— UNDER GRADUATE STUDIES DIVISION

FIRST SEMESTER 2026-27

Course Handout Part II

Date: 01.08.2026

In addition to part I (General Handout for all courses appended to the Time table) this portion gives further specific details regarding the course.

Course No : PHA F242
Course Title : Biological Chemistry
Instructor-in-Charge : Richa Shrivastava, richa.shrivastava@pilani.bits-pilani.ac.in, 3170-B
Practical Instructors : Jyotika, Trivedi Mansi Vinodkumar, Rakshe Pratik Shankar, Jyoti Verma, Kulkarni Hrushikesh Vikas

1. Course Description: This course focuses upon the physiological and medical significance of important biomolecules along with their use as drug targets. The course also elaborates the role of vitamins as prosthetic groups of enzymes and enzyme kinetics and regulation. The course deals with the understanding of chemical structures, biochemical reactions metabolic pathways related to carbohydrates, lipids, nucleic acids and protein synthesis and diseases arising from defects in these pathways.

2. Scope and Objective of the Course: The aim of this course is to describe and explain all biochemical processes of living organisms and their interactions with their networking both in health and disease conditions.

3. Course Learning Outcomes: Understanding of basic metabolic pathways, diseases associated with errors in the pathways and drug targeting.

4. Text Books and references*:

Text Books: Robert K. Murray, et.al, Harper's Illustrated Biochemistry, McGraw Hill Medical Publishers, 29th edition, 2012

Reference Books: David L. Nelson, Michael M. Cox. Lehninger Principles of Biochemistry, W.H. Freeman Publishers, 7th edition, 2017

***Apart from text books and reference books refer class notes**

5. Course Format and weekly workflow

6. Course Plan:

WK No.	Module Number	Lecture Session	Reference	Learning Outcome
1.	1. Overview of Biochemistry	1. Introduction to Biochemistry and its relation to medicine (Lecture 1)	T Ch. 1: R.a. Ch. 1	Exposure to the general biological chemistry and its targeting in health and medicine
1.	2.Introduction to Carbohydrates	1. Carbohydrates: Classification, Glycosidic bond formation, Detection test, properties and stereochemistry.	T Ch. 14: R.a. Ch.7	Physiological and Medical Significance of important Carbohydrates .



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		(lecture 2-4)		
2-4	3. Carbohydrate Metabolism	1. Glycolysis and the oxidation of pyruvate (Lecture 5-6) 2. The Citric acid cycle: The catabolism of Acetyl CoA (lecture 7) 3. The Pentose phosphate pathway(lecture 8) 4. Glycogen metabolism (lecture 9-10) 5. Biomedical significance (lecture 11)	T Ch. 18: R.a. Ch.14 T Ch. 17: R.a. Ch. 16 T Ch. 21: R.a. Ch. 14 T Ch. 19	Complete overview of carbohydrate metabolism with emphasis on the metabolic defects involved in diseases like Diabetes mellitus, glycogen storages diseases, Wernicke-Korsakoff Syndrome G6PD Deficiency: Anemia, Lactose intolerance, Galactosemia etc.
5	4. Oxidative Phosphorylation	1. Electron transport chain 2. ATP synthesis 3. Regulation of Oxidative phosphorylation (lecture 12-13)	T Ch. 13 R.a Ch. 19	Will help in understanding of as how different metabolic pathways merge at electron transport chain and cause ATP synthesis in mitochondria. Mechanism of action of drugs like cyanide.
5-7	5. Vitamins and Coenzymes	1. Classification of Vitamins (lecture 14) 2. Structure and functions of some important vitamins as coenzymes (lecture 15-18) 3. Deficiency disorders and therapeutic and diagnostic applications of enzymes and isoenzymes (lecture 19)	Class notes. References/sources will be given in slides.	Students will acquire basics information of various aspects of vitamins. How various enzymes work with the help of vitamins as prosthetic groups.
7-9	6. Enzymes	1. Classification and mechanism of action (lecture 20-21) 2. Enzyme kinetics (lecture 22)	T Ch. 7,8,9 R.a. Ch. 6	Detailed understanding of structure and functions of enzymes. Understanding



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		3. Enzyme: Regulation of activities (lecture 23-24)		their mechanism of action, regulation and enzyme kinetics. Discussion on the role of enzymes in diseases, as biochemical tools and drug targeting.
11-12	7. Lipid metabolism	1. Introduction to Lipids (Lecture 25) 2. Transport and storage (lecture 26) 3. Oxidation of fatty acids (lecture 27-28) 4. Biosynthesis of fatty acids (lecture 29-30) 5. Biomedical significance (lecture 31)	T Ch. 15: R.a. Ch. 10 T Ch. 22: R.a. Ch.17 T Ch. 22: R.a. Ch.17 T Ch. 23: R.a. Ch.17	Basics and nomenclature of lipids. Emphasis on Omega fatty acids. Knowledge of various aspects of lipid metabolism. Defects in lipid metabolism leading to diseases like atherosclerosis and obesity, zellweger, ketosis etc. Drug targeting using HMG-CoA enzyme. Sex hormones and contraceptives.
13-14	8. Amino acid and protein metabolism	1. Structure of amino acids and proteins (lecture 32) 2. Catabolism of amino acid nitrogen (lecture 33) 3. Catabolism of carbon skeleton of amino acids and their Conversion of amino acids to specialized products, Such as Porphyrins and bile pigments (lecture 34-35) 4. Biomedical significance (lecture 36)	T Ch. 3,4,5: R.a.Ch. 3 T Ch. 28: R.a. Ch.18 T Ch. 29: R.a. Ch.18	Amino acids and structure of proteins. Exposure to protein metabolism. Role of amino acids as blood buffers against metabolic acidosis. Defects in nitrogen metabolism leading to Phenylketouria and Alkaptonuria etc.



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14-15	9. Nucleic acids and their metabolism	1. Nucleic acids (Lecture 37) 2. Metabolism of purine and pyrimidine nucleotides (lecture 38-39) 3. Biomedical significance (lecture 40) 4. DNA Replication (lecture 41) 5. Transcription or RNA synthesis ((lecture 42)	T Ch. 32: R.a. Ch.8 T Ch. 33 R.a. 25,26,28	Detailed study of synthesis and degradation of nucleic acid. Therapeutic targeting of nucleotide metabolism in diseases such as Gout and cancer.
15	9. Introduction to endocrinology	Classification and functions of important hormones (lecture 43)	T Ch. 41.	Overview of the various roles of hormones

6. Laboratory Practical List:

1. Introduction to biochemistry practical: Risk involved while handling chemicals, safety precautions that are necessary while working in lab, marks distribution for lab component.
2. Identification of Carbohydrate-I
3. Identification of Carbohydrate-II
4. Quantification of Vitamin C in the sample
5. Identification of Lipids
6. Identification of Proteins
7. Urine Analysis
8. Protein estimation.
9. Enzyme catalysis: Videos and numerical
10. Separation techniques for proteins
- 11-12. Chromatography: Paper and TLC

7. Evaluation Scheme:

Component	Weightage (%)	Duration	Date & Time	Nature of component (Close Book/ Open Book)
Mid-Semester Test	25	90 Min.	<TEST_1>	Closed Book
Comprehensive Examination	40	3 h	<TEST_C>	Closed or/and Open Book



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Lab Component	25			
Surprise quiz(es)	10	Continuous		Closed Book

Regularity in attendance will be one of the criteria in deciding the borderline cases at the time of final grading as well as make-up's.

It is not necessary that all the grades (i.e. A to E) would be awarded.

In borderline cases subjective judgment will be exercised for pull-up's (max. 2%). Basic guiding factors will be regularity, consistency in performance (above average) or/and steady improvement throughout the semester. Students scoring less than 20% will be considered NOT exposed to the subject.

8. Chamber Consultation Hour: 5:00 PM-6:00 PM [Email: richa.shrivastava@pilani.bits-pilani.ac.in]

9. Notices: Notices pertaining to this course will be sent via emails.

10. Make-up Policy: Make-Ups are not given as a routine. It is solely dependent upon the genuineness of the circumstances under which a student fails to appear in a scheduled evaluation component. In such circumstances, prior permission should be obtained from the Instructor-in-Charge. The decision of the Instructor-in-Charge in the above matter will be final. There is no makeup for continuous evaluation.

11. Note (if any):

Instructor-in-charge
Course No. PHA F242